

The Old and the New Axioms of User Interfaces

Unspoken Existence, Crisis, and Implications for Design and Engineering

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ABSTRACT

Axioms are self-evident propositions used in formal deductive systems. This work intends to show that there are unrecognized axioms of user interface design that serve as starting points for building interactive systems. From foundational theories, laws, principles, and guidelines, and through a distillation process, the identified axioms are enunciated and explained. The set of five is called the old axioms and they are in crisis. Old axioms are insufficient to support new developments and trends in Human-Computer Interaction. A new set is needed, and it is proposed here. The five new axioms shift focus from “computer-centric” to “object-of-interest-centric”. Nowadays, computing devices come and go, but the digital objects that belong to users, remain. That is why the center of attention has to move and adhere to new principles.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); HCI theory, concepts and models; Interaction design; Interaction design theory, concepts and paradigms; Collaborative and social computing; Collaborative and social computing theory, concepts and paradigms.

KEYWORDS

User Interface, Axioms of User Interfaces, Old Axioms, New Axioms

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1 INTRODUCTION

An axiom is a self-evident truth that requires no demonstration [9]. If its validity is questioned, it is usually enough to analyze the absurd consequences of assuming its negation. Axioms are commonplace in fields like mathematics and logic, but in no way exclusive to those. They provide a solid foundation for deduction and proof and in consequence, are highly appreciated and carefully formulated by scholars that seek the formalism and certainty of deductive procedures. Computer Scientists use them too, but if we turn to HCI literature, it will be difficult to find examples. The empirical

nature of HCI methods seems to discourage their use, suggesting inductive or abductive studies instead. It does not mean that axioms are futile in HCI. This work pretends to show that there are axioms in place, maybe not explicitly enunciated but undoubtedly present.

In this paper, some axioms will be enunciated in direct connection with user interfaces, their design and usability. If done correctly, enunciating them should be enough to recognize their validity. This means that they should make sense by themselves, with minimal argumentative support. After learning them, the reader could acknowledge that they have been in use as starting points for deducing the general structure of UIs, and problems arise that affect usability when neglected.

UIs are objects of study in HCI, but HCI is not limited to UIs. The axioms presented here belong to the realm of UIs and their design. HCI axioms exist as well, as mentioned in [42] and [12], and attempts to formulate them for software development can also be found (see [21]).

Axioms can take different forms. The mathematical style of writing axioms employs symbols and notations that are almost ambiguity-free. There is also a colloquial style, that does not require special notation or symbols, just everyday language, but needs careful wording to avoid misinterpretations or confusion. The colloquial style, employed here, seems more adequate for a domain in which conceptual exchanges among stakeholders, like researchers, designers, users, engineers, students, and others, are common because it uses terms that are understandable to all.

This paper does more than just present a set of axioms and make a case for their validity and actual influence in UIs, even if they have never been enunciated before. This initial set is defined as the “old” axioms. Old axioms have been useful and productive, but we may be approaching a limit to their usefulness. The case could be analogous to that of ancient astronomical axioms. They worked fine and helped advance science and technology up to a point where reformulation and change were needed. The advances in hardware and software have resulted in considerable increments in the computational power available. UIs could now be the bottleneck that prevents users from taking full advantage of that power. The “new” axioms could reframe our understanding of UIs and open new paths for the design and development of interactive systems, but only if we agree on them.

A word of caution before continuing. HCI is a practical field in which problems demand solutions urgently. Delivering a product that interacts efficiently with its intended users seems much more valuable than developing theories or axiomatic frameworks. Models and theories exist, only that they advance at a slower pace, in some cases with years or decades between one step and the next. The collateral effect is that some references and seminal work for papers like this appear old and outdated when compared with what

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is found in material related to current technological trends where research work is vibrant and updated frequently. It is like, regarding foundational theories and conceptual frameworks, we are having a slow conversation that takes long breaks between interventions.

2 DISCOVERING AXIOMS

Axioms cannot be invented, only discovered through study, reflection, and introspection. Starting with knowledge taken for granted, we proceed backward questioning why that knowledge is certain and from which premises it derives. Quite often, certainty is so strong that no one feels the need to question and start that seek process. It happens in HCI and in other fields where axioms remain implicit but never identified. In those cases, a specific intention to look for them must be formed and justified. That is why some context is needed here.

I started my research project about UIs in 1990 when I noticed that the HCI field had supportive theories for computers and users but not for the bridge between them. The works of Alan Turing, Claude Shannon, and Ludwig Von Bertalanffy, to name a few, provide foundations for computer science and artificial intelligence, information theory, systems theory, and cybernetics [39]. The multidisciplinary nature of HCI rests in other fields to learn about users, including psychology, anthropology, philosophy, sociology, ergonomics, management, and others [14, 27, 37].

In the case of user interfaces, HCI has made attempts to advance foundational theories on its own. For example, the works of Michel Beaudouin-Lafon, although he argues for better models and theories of interaction and not of user interfaces, and concludes that “HCI is not the science of user interfaces, just as astronomy is not the science of telescopes” [5], he clearly explains the benefits of interaction models reasoning that they “provide a framework for guiding designers, developers and even users (in the context of participatory design) to create interactive systems” [5]. His proposal of instrumental interaction [4] anticipates the need to transcend current interaction design systems. Also, along with Wendy Mackay, they propose three general principles for designing visual interfaces: reification, polymorphism, and reuse [6].

Other approximations include Bob Baxley’s universal model of a user interface [3], Jesse James Garrett’s elements of user experience [19], Jan Gulliksen and coauthors’ key principles for user-centered systems design [18], Jakob Nielsen’s usability heuristics [31], Robert Jacob and coauthors’ reality-based interaction [22], Jonathan Grudin’s levels of development of UIs [16], Jef Raskin’s laws of user interface design [35], and profound propositions as those of Gerrit van der Veer and Hans van Vliet that state that from the user’s perspective, the human-computer interface is the system [40], or Thomas Moran’s declaration about UI design in his 1981 paper “it should be efficient to use and easy to learn; it should be consistent, logical, ‘natural’; amen. There are at present no systematic methods or body of knowledge to help the designer accomplish these goals for the interface—it is a purely intuitive endeavor.” [29] To a certain extent, this last claim is still valid. In fact, several authors have categorized UI design as an art rather than a technical endeavor or systematic practice (see [8, 26]).

Once and again, HCI literature has recognized that deficiency. Kasper Hornbæk et al., made a thorough review of 35 years of

CHI publications and in their report, they say that “the notion of interaction is essential to human-computer interaction, yet rarely studied” [20]. Others have attempted similar efforts, searching the literature across time and looking for foundational knowledge (see [13, 24]).

Lars-Erik Janlert and Erik Stolterman write, “Human-computer interaction (HCI) used to be about interfaces and their design. Rarely, however, has the field devoted any serious time and energy to a reflective examination of what an interface is or how it could or should be defined” [23]. In their work, they present 4 instances of what they call “thought styles” about interfaces in general, or how the interface can be thought of as a surface of contact between matching objects, a boundary of an independent object, a means of controlling an object, and a means of expressions and impressions, a target of interpretations and affectations.

Simon Herbert’s work developing sciences of the artificial provides relevant insights for UIs and HCI. Among them, the idea that “an artifact can be thought of as a meeting point an “interface” in today’s terms between an “inner” environment, the substance and organization of the artifact itself, and an “outer” environment, the surroundings in which it operates.” [38] Since interactive systems are artifacts, his conceptual framework is useful and worth considering.

Pure axiomatic methods for specifying the semantics of user interfaces are documented in HCI literature. In a 1985 paper, Uli Chi presents four axiomatic approaches and applies them to the development of the user interface of a commercial product [11]. In that work, axioms are not intended to be general or applicable to other systems. They are domain-specific enunciations stated in such a way that they become the starting point for the interpretation of functions in that domain. Developers use these specifications to build the system. It can be conjectured that extending and generalizing that branch of research could have led to the identification of universal axioms.

Axioms must be distilled from that conceptual background. To a certain extent, I can affirm that was the process I followed, although I am unsure if I can thoroughly describe it. Constantly reviewing guidelines, laws, principles, heuristics, theories, models, definitions, and frameworks, may lead to discovering underlying concepts that can be generalized as axioms. The soundness of them should supplement further analysis and even explain their origin.

3 THE OLD AXIOMS

Axioms that have been in place for long, exercising their role as initial postulates, can be labeled as old, and that does not mean that they have been discarded or refuted, instead, this adjective emphasizes the solid position they have held and still hold. The axioms enunciated here belong to the domain of design and construction of UIs.

Regarding axiomatic systems, Kurt Gödel’s first incompleteness theorem states that there will be true statements within the system that cannot be proved nor disproved from the system’s axioms [34]. The practical implication of this theorem for UIs is that the old axioms will become insufficient to account for all principles, heuristics, models, theories, and frameworks that are in fact (truly) useful within the domain of UI design. Hence, new axioms need

to be discovered, only to become insufficient again with time. But new axioms cannot be postulated without first enunciating the old ones.

What follows is the enunciation of the set of old axioms. I will codify each one as OA, for Old Axiom, and a sequence number, starting with 0.

3.1 Old axiom 0 – OA0 fundamental axiom.

Axiom statement: Computers work by performing functions.

OA0 states that the work computer systems can do is provided in the form of discrete units that execute a particular function, taking inputs and internal states to work on, and producing outputs. Computers advance when they offer new functions, coherently integrate separate functions into meta-functions, execute functions more efficiently, or become capable of handling more inputs, state variables, and outputs at the same time. There is no other type of work that computers can perform.

3.2 Old axiom 1 – OA1 generative axiom.

Axiom statement: Every computer function that produces useful output for a user must have mechanisms of activation and delivery of results.

OA1 is called generative axiom because it generates the need for user interfaces. Activation mechanisms collect input and issue the command for executing a function. Delivery mechanisms present the output of the function to the user. There are no computers with useful functions without activation and delivery mechanisms. The evolution of devices used for activation and delivery has gone from switches, knobs, dials, cable connections, punched cards, analog gauges, indicator lights, printers, and teletypes in the first computers [17, 27] to modern displays and virtual reality devices, sound and scent synthesizers, brain electromagnetic signals detectors, and combinations of them in recent projects [1].

3.3 Old axiom 2 – OA2 expansion and reuse axiom.

Axiom statement: The number of functions in computers grows at a rate that is greater than the rate of growth of the number of activation and delivery mechanisms.

OA2 describes the relationship between growth rates of computer functions and activation and delivery mechanisms across time. One of them is greater than the other. OA2 implies that as functionality expands, some activation and delivery mechanisms will have to be reused or grouped, through menus or modes of use, for example, to maintain consistency with OA1.

With computer advances, UI designers realized that adding a new device to activate and deliver results of new functions was not practical. Instead, reusing what was available was much more practical, especially because new functionality relied on software while devices involved hardware. Software is much easier to distribute, maintain, and repair than hardware. The result has been that input and output devices have been reduced to a stable set of components.

3.4 Old axiom 3 – OA3 limitation axiom.

Axiom statement: Human capability imposes a limit to the number of functions a computer can effectively provide.

OA3 anticipates what could happen if the number of functions in a computer exceeds a threshold that depends on human capabilities. Beyond that threshold, additional functionality will become invisible to the user, even if activation and delivery mechanisms are provided and, hence, futile, or inaccessible. OA3 imposes the need to look for packaging functions to reduce the actual offer to the user.

There are functions in computers, think of setting a byte size memory position to a certain value for example, that are so basic, that they are buried inside macro functions or packages that implement a higher-level function. This does not contradict OA1 because basic functions are activated indirectly and, under certain conditions, can be activated directly and their results delivered immediately, as in the case of debugging a program in assembly language.

3.5 Old axiom 4 – OA4 imitation axiom.

Axiom statement: Configurations of activation and delivery mechanisms become popular or de facto standards without implying that they are the best choice for users of that functionality.

OA4 makes a point regarding the selection of configurations of activation and delivery mechanisms in a user interface. Designers need not look for the best alternative for users if they can identify the de-facto standard in similar systems. This dynamic creates imitation trends that benefit designers and users. A popular example is design patterns [41].

3.6 Old axioms workflow

The set of axioms from OA0 to OA4 are combined in the mind of engineers and designers who build interactive systems, following a line of thought that goes like this: Here are some computer functions, existent or planned, (AO0), whose users need activation and results' delivery mechanisms (AO1), that functionality will be added on top of existing capabilities and no increment in the number of activation or delivery mechanisms will occur, hence some rearrangement or accommodation must be done (AO2), doing it in a way that puts the least stress in human abilities like attention, memory, learning, synthesis, analysis, and others, that can be surpassed if a certain threshold is passed (AO3), and to do it safely de facto standards for that class of functions should be observed (AO4).

4 AXIOMS IN CRISIS

At this point, the set of axioms presented may appear “computer-centric”. It starts with computer functionality (OA0) and proceeds outwards considering how those functions can be made available to users (OA1 to OA4). Modern interactive systems are less computer-centric than they were at the beginning of the information age. Several trends challenge this paradigm and require a change of perspective.

Distributed computing [25, 43] is one of them. If functions are provided by different interconnected systems that can be geographically near or distant from each other, then the actual number of

functions available is unknown in advance and UI design has to proceed with a certain level of blindness, and instead of thinking of activation and delivery, coordination and request of functionality must be addressed.

Social computing [33, 36] is another. Information objects, sometimes known as “content”, are created collaboratively by groups of individuals each one contributing their skills and resources. Depending on the robustness of the links between individuals in the group, it could be a team, a community, or a mass of users. Interactive systems for social computing are more focused on providing means of collaboration and sharing of objects of interest among participants.

Other forces driving change are multimodality [28, 32] in which multiple communication channels are open to the exchange of information between interactive systems and users, and, importantly, interactive systems can pay attention to human sentiments and feelings by combining input from voice, gestures, typing, facial expressions, and others; and the diversity of devices available for use or cross-device interaction [10, 30] where users expect to access their content or objects of interest from different devices that function as a sort of terminals that can be replaced as needed. Information technology moved from the impersonal paradigm of mainframes to the personal computers of the last decades of the 20th century, and now it looks like a fractioned landscape where users deposit all kinds of portions of their personality, from the very intimate to the most public, with support for all forms of human relationships, commercial, romantic, professional, formative, and spiritual.

The computer-centric nature of old axioms does not fit well with these new trends and forces. In line with the transition across the three waves of HCI [15] that provides a complementary vision of the problem, it is time to propose new ones.

5 APPROACHING NEW AXIOMS

Before embarking on a lengthy discussion about a proposal of new axioms, I should first discuss why the proposal must be short and minimal instead. Following Ockham’s razor [2] the unnecessary multiplication of entities, axioms in this case, would only lead to confusion and the risk of signaling in the incorrect direction. Proposing an axiom that suggests developments in a branch of research may be misleading and superfluous. Only the basic axioms should be proposed now. The rest should be enunciated as the field advances.

The basic shift proposed now is from “computer-centric” to “content-centric” or “object-of-interest-centric”. Some might suggest that the shift should be to “user-centric”. I would argue that that shift has already been done. We cannot make systems more friendly or improve user experiences in using interactive systems until we change the design focus. The alternative is then between content (for content-centric axioms) or object-of-interest (for object-of-interest-centric axioms). I believe content is too general and assumes information can be separated from the objects to which it belongs. Content has become a popular term because it emphasizes the shareable data that is curated and presented through UIs, but it obscures the fact that data is nothing without an object of reference.

Focus on objects opens the road to information, knowledge, and wisdom [7].

Hence, I believe the approach to new axioms should be “object-of-interest-centric”. The inclusion of the “interest” part of the phrase makes direct reference to human users. Machines do not have interests, only humans do. Personal interests have a story behind them and are not generic, they emphasize individuality, personality, and future trajectory.

6 THE NEW AXIOMS

The use of the adjective new for the axioms that follow indicates that they did not have a role in the initial phases of modern computer technology. There was a time in which computers were neither personal nor ubiquitous. Much less were computers interchangeable with other smart devices, even disposable. Axioms related to those features, or their implications were unnecessary, even confusing. Because it was not possible to build the interactive systems where they apply, their enunciation had to wait. With advances and infrastructure development, an intuition of them has started to form. As with the old axioms, this new set will become insufficient to accommodate new developments.

The proposed new axioms will be articulated in the following paragraphs using a similar codification as the old ones. NA, for New Axiom, followed by a sequence number starting with 0. I will use ObIn, plural ObIns, to refer to objects of interest, which are those that users want to manage and maintain using computers or smart devices, usually called interactive systems, information systems, or just systems. ObIns can be files, designs, contacts, posts on social media, different forms of code, physical objects, or other content including those that receive feedback from acquaintances or the public.

6.1 New axiom 0 – NA0 object ownership axiom.

Axiom statement: Users have objects of interest (ObIns); they own and share them.

NA0 extends the concept of ObIn with two new features, ownership and sharing. Interest in an object develops from within a person or group because of a connection with it that the person acknowledges. That connection originates from character and personality and is expressed as interest. Strong interest leads to desires of possession or ownership, be it exclusive or shared with others. Ownership implies that users want to access and use an ObIn across time and devices and control the extent to which other users access it.

6.2 New axiom 1 – NA1 systems sustain axiom.

Axiom statement: Users employ systems to help sustain the existence of their ObIns.

NA1 expresses the need of users to employ systems, mostly information systems, to sustain the ownership of their objects of interest. Users seek reassurance of the relationship between them and their objects, their current states, degree of progress towards a goal, sharing status, and other features.

6.3 New axiom 2 – NA2 representation axiom.

Axiom statement: Systems provide representations of ObIns not the actual ObIns.

The concept of ObIn can be seen as a replacement for the notion of file or document stored in a computer. The comparison can play the role of a useful metaphor. However, an ObIn is much more complex than a file or document. A file stores content, that is data. An object can be something tangible and physical, or a creation in the mind of the user, with a structure and conceptual unity. Hence, ObIns cannot be stored in a system, only their representations. A representation is an artifact that makes the object present again, re-presents the ObIn, which means it must convey information, relations, knowledge, history, and condition of the ObIn. Also, the representation should allow work and analysis on the actual ObIn.

6.4 New axiom 3 – NA3 basic operations axiom.

Axiom statement: Systems enable users to create, capture, update, and delete representations of their ObIns.

NA3 describes the basic operations that are possible with ObIns. A user may have an idea of an object that has never existed, a new form of vehicle, furniture, dish, artifact, essay, or any other thing. The user creates that object in the mind, and that same process should be possible in systems. If the object already exists physically, then its representation must be initially brought to a system, that would be a capture operation. The evolution of the ObIn should be reflected in its representation through an update operation across systems. Finally, when the user no longer wishes to have that ObIn then its representation should disappear through a delete operation.

The operation of creation correlates with or even causes the ownership of the ObIn. There is ample room here for discussion, especially if the other three basic operations are also considered, but I will leave that topic for future work.

6.5 New axiom 4 – NA4 share axiom.

Axiom statement: Systems enable sharing ObIns among communities of users.

NA4 recognizes the modern dynamics of work and creativity in the interconnected world. Users share their creations and spread ownership among members of teams and communities, fostering respect and observation of boundaries of influence. Collaboration enhances the products by activating different skills or sets of resources.

As mentioned before, this set could be extended with axioms that account for access methods, search possibilities, and edition of features of ObIns. The point of this paper is that the new set can be proposed and analyzed. Future work will complement this, hopefully with contributions from other researchers.

6.6 New axioms workflow

Work with the new axioms could follow a pattern as follows: a designer identifies a class of ObIns that is owned and shared by a community of users (NA0). Recognizing that class of ObIns is not currently supported by systems, she designs a workflow addressing the particular form in which this class of ObIns is owned and shared by that community (NA1). A fundamental portion of that design specification is the way in which the ObIns will be represented in

the system (NA2). When the representation has been codesigned with the community of users, the mechanisms to create, capture, update, and delete representations are designed (NA3). The sharing procedures (NA4) are set in accordance.

The contrast between the workflow with the old and new axioms is stark. The main difference is the starting point. With the old axioms, the designer considers first the functions to be offered to the user. In the new ones, the ObIns, which express the interests of users, are most prominent. However, there is room for applying the old axioms in conjunction with the new ones, in a sort of embedding, in the design portion where available functionality has to be made accessible to the user for the set of ObIns under consideration. After the representation (NA2), and basic operations (NA3) have been designed, materializing those designs in specific devices or platforms requires considering the functionality they provide and how it is offered to the user, which invokes the use of OA0 to OA4.

7 AXIOMS SHIFT

A comparison between the old and the new axioms should make patent the difference in focus. Old axioms are “computer-centric” whilst new axioms are “object-of-interest-centric”. An important feature of ObIns is that they are application-independent, meaning that it does not matter who created one, any user could contribute useful work to enhance it with the skills and resources at hand.

Although the shift is evident, we could question if it was the right shift to take, if users, designers, and engineers are willing to adopt these axioms, or if the technology is ready to support them. An interesting thing about this process is that, in my opinion, it cannot be made unintentionally. If a fundamental change is to be made in our conceptual model of information systems and if we expect that change to lead design, construction, and use of systems in accordance with that new paradigm, then it must be intentionally adopted. That means, it has to be proposed, evaluated, accorded, and executed.

The old axioms are highly ingrained in our practices and concepts. There is no apparent reason for them to vanish in favor of better practices. A change might not occur spontaneously.

8 CONCLUSION

The axiomatic approach is not common in the field of HCI. However, in this paper I discussed how there have been advances in foundational theories and models from which it is possible to distill underlying axioms, although it may not be evident how to achieve their proper enunciation.

Assuming the risk of making mistakes, I proposed the enunciation of old axioms, remarking that the adjective old is used to emphasize their presence from the beginning of the computer era, and not to indicate that they are obsolete or disproved. The list of old axioms follows.

- OA0 fundamental axiom.
- OA1 generative axiom.
- OA2 expansion and reuse axiom.
- OA3 limitation axiom.
- OA4 imitation axiom.

A case for the need to propose new axioms was presented with reflections of caution. At this moment it may not be effective to

advance a long list of new axioms. It is better to enunciate a few and add new ones in future works. The list of new axioms follows.

- NA0 object ownership axiom.
- NA1 systems sustain axiom.
- NA2 representation axiom.
- NA3 basic operations axiom.
- NA4 share axiom.

A clear shift in the design, construction, and use of computers and systems was presented.

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